Double-Slit Experiment: Wavelength Determination of Laser Light

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INTRODUCTION

Before the 20th century, scientists had a long history of attempting to understand the nature of light. Issac Newton proposed the corpuscular theory, that light is made of tiny particles called corpuscles. According to Newton, light travels in straight lines at high speeds, and this theory attempted to explain how particles reflect off surfaces and refract in different media. Thomas Young demonstrated that light behaves as a wave in his groundbreaking double-slit experiment, making the two theories seem fundamentally incompatible. The rise of quantum mechanics resolved this paradox, revealing that light can exhibit both wave and particle nature depending on how it is observed.

The double-slit experiment is still one of the most influential experiments that demonstrates quantum mechanical principles. When a coherent light, like a laser, shines through two close, narrow slits, a pattern of light is projected on the surface behind the slits. This pattern of alternating light and dark bands could only be the result of a wave nature, despite light being only detected as particles known as photons. This dual nature of photons led to the breakthrough that this principle extends to all quantum particles.

The interference pattern resulting from the double-slit experiment provides a method for calculating the wavelength of light. The spaces in the resulting pattern depend on the wavelength and slit size. In this experiment, red and green lasers were used to create interference patterns at different slit sizes and multiple distances. Measuring the resulting pattern and applying error analysis enabled the determination of wavelengths and comparison with their rated values.

THEORY

When monochromatic light, or light that is at a single focused wavelength, passes through two closely spaced, narrow slits, it diffracts and creates two light sources. The resulting waves overlap and produce a pattern of bright and dark regions on the screen, some distance d from the slits. The maxima occur when the waves are overlapping in phase and constructively interfere, whereas the minima occur when the waves are out of phase and destructively interfere.

These resulting bands can be predicted using the path difference between the light traveling from each slit to a point on a screen. This can be represented with the following equation:

Equation 1: Double-slit Interference Equation

$$\lambda = \frac{xd}{(nL)}$$

In which λ is the wavelength of the light, x is the distance from the central band to another band (n orders away), d is the separation between the two slits, n is the number of bands, and L is the distance from the slits to the viewing screen.

The central band (n = 0) will be the brightest band located at the center of the pattern, this is where the path difference is zero. Since the pattern is symmetrical, moving in either direction results in the band order increasing by the same amount (n =1,2,3…). Through the measurement of the central band and the distance between the bands, the wavelength of the light source can be determined. The slit width only affects the intensity of the light in the distribution, but it does not affect the band spacing, which is why it is not included in the calculations.

EXPERIMENTAL

Equipment

The experiment utilized Class II red and green lasers with an output power of less than 1 mW. A tool with adjustable slit widths, specifically utilizing the slit separations of d = 0.25mm and d = 0.5 mm. Both patterns used the same slit widths of 0.04mm. White paper was used as a viewing screen and marked to visualize the patterns, then measured with a ruler to get accurate band positions.

Experimental Procedure:

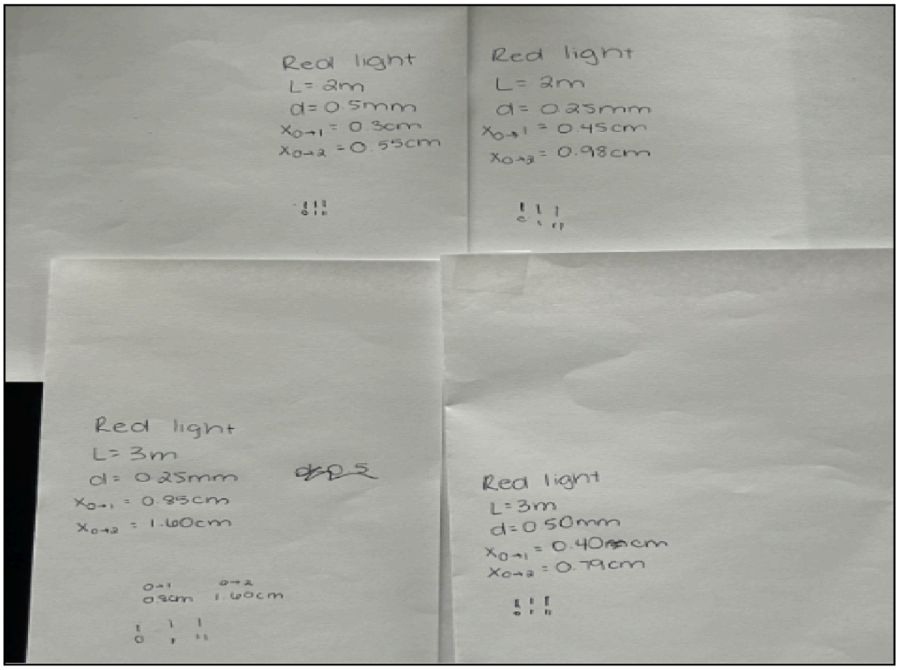
The slit tool was positioned 10cm in front of the laser, aligned so the beam perpendicularly passed through the center of the double-slit pattern. Trials were conducted at two distances: 3 m and 2 m between the slit tool and the viewing screen. These distances were measured using meter sticks.

A white piece of paper was taped to a cardboard box and served as the viewing screen. Once the interference pattern was visible, a pencil was used to mark the lines through the center of each band. The central band was identified through the use of symmetry of the pattern. Bands were counted on either side to verify their position.

The experiment was performed using two slit separations ($d = 0.25$ mm and $d = 0.5$ mm) and two screen distances (3 m and 2 m). For each of the four trials, measurements were taken at two bands ($n = 1$ and $n = 2$), resulting in eight measurements. After the bands were marked, the distances between the outer bands and the central band were measured with a ruler for calculations.

DATA

Figure 1: Marked Interference Patterns for Red Laser

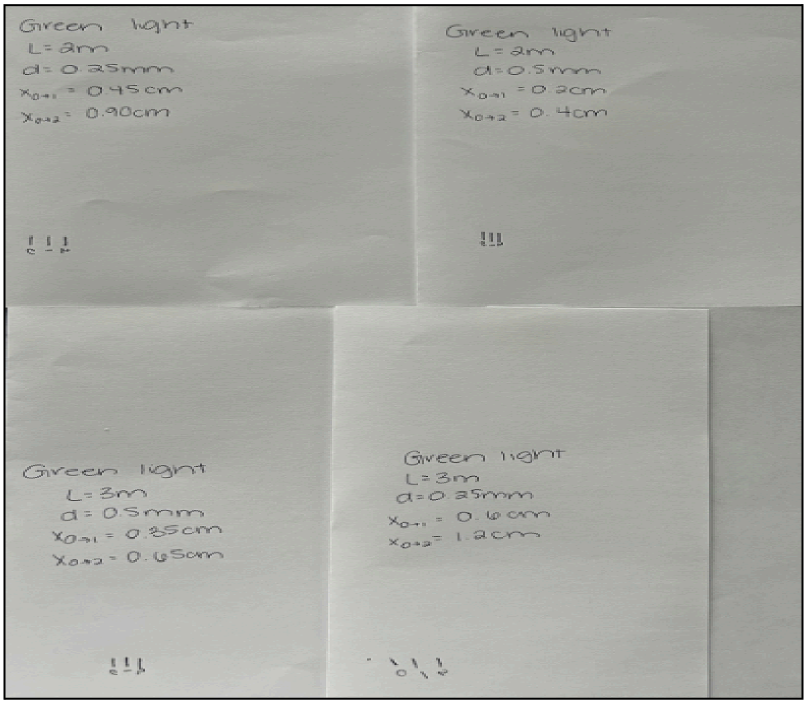


This figure shows the marked interference patterns for the red laser at different experimental configurations. Each paper is marked with distance to the screen (L), slit separation (d), and measured distances from the central to the first and second bands (x_{0-1}, x_{0-2}).

Table 1: Measurements and Calculated Wavelengths for Red Laser

d (mm)	d (m)	Length (m)	n	x (cm)	x (m)	λ (m)	λ (nm)
0.25	0.00025	3	1	0.85	0.0085	7.0833e-07	708.33
0.25	0.00025	3	2	1.60	0.0160	6.6667e-07	666.67
0.50	0.00050	3	1	0.40	0.0040	6.6667e-07	666.67
0.50	0.00050	3	2	0.78	0.0078	6.5000e-07	650.00
0.25	0.00025	2	1	0.45	0.0045	5.6250e-07	562.50
0.25	0.00025	2	2	0.98	0.0098	6.1250e-07	612.50
0.50	0.00050	2	1	0.30	0.0030	7.5000e-07	750.00
0.50	0.00050	2	2	0.55	0.0050	6.8750e-07	687.50

Figure 2: Marked Interference Patterns for Green Laser



This figure shows the green laser interference patterns with the same parameters as the red laser.

Table 2: Measurements and Calculated Wavelengths for Green Laser

d (mm)	d (m)	Length (m)	n	x (cm)	x (m)	λ (m)	λ (nm)
0.25	0.00025	3	1	0.60	0.0060	5.0000e-7	500.00
0.25	0.00025	3	2	1.20	0.0120	5.0000e-7	500.00
0.50	0.00050	3	1	0.35	0.0035	5.8333e-7	583.33
0.50	0.00050	3	2	0.65	0.0065	5.4167e-7	541.67
0.25	0.00025	2	1	0.45	0.0045	5.6250e-7	562.50
0.25	0.00025	2	2	0.90	0.0090	5.6250e-7	562.50
0.50	0.00050	2	1	0.20	0.0020	5.0000e-7	500.00
0.50	0.00050	2	2	0.40	0.0040	5.0000e-7	500.00

Table 3: Average Wavelengths Compared to Known Values

Laser Color	Average λ (m)	Average λ (nm)	Standard Deviation (nm)
Red	6.6406e-07	664	56
Green	5.3125e-07	531	32

CALCULATIONS

Calculation 1: Sample Wavelength Calculation for Red Laser

Using the double-slit interference equation:

$$\lambda = \frac{xd}{(nL)}$$

For the red laser at L = 3m, d = 0.25, n = 1, x = 0.85cm

$$\lambda = \frac{(0.0085\, m)(0.00025\, m)}{(1)(3\, m)}$$

$$\lambda = 7.0833 * 10^{-7} m = 708.33\, nm$$

Calculation 2: Average Wavelength for Red Laser

$$\lambda_{RedAverage} = \frac{(708.33+666.67+666.67+650.00+562.50+612.50+750.00+687.50)}{8} = 664.06\, nm$$

Calculation 3: Average Wavelength for Green Laser

$$\lambda_{GreenAverage} = \frac{(500.00+500.00+583.33+541.67+562.50+562.50+500.00+500.00)}{8} = 531.25\, nm$$

Calculation 4: Percent Difference Between Experimental and True Value for Red Laser

True value for red laser is 650nm:

$$\%\, Difference = \frac{|Experimental - Known|}{Known} * 100\%$$

$$\%\, Difference = \frac{|664.06nm - 650nm|}{650nm} * 100\%$$

$$\%\, Difference = 2.16\%$$

Calculation 5: Percent Difference Between Experimental and True Value for Green Laser

True value for green laser is 532nm:

$$\% \, Difference = \frac{|Experimental - Known|}{Known} * 100\%$$

$$\% \, Difference = \frac{|531.25nm - 532nm|}{532nm} * 100\%$$

$$\% \, Difference = 0.14\%$$

Calculation 6: Error propagation for red laser at L = 3 m

For the double-slit interference equation: $\lambda = \frac{xd}{(nL)}$

Given the Uncertainties:

$\Delta x = 0.1 \, mm = 0.0001 \, m$

$\Delta d = 0.01 \, mm = 0.00001 \, m$

$\Delta L = 1 \, mm = 0.001 \, m$

Using the trial: L = 3 m, n = 1, x = 0.0085 m, d = 0.000250 m

The error propagation for $\lambda = \frac{xd}{(nL)}$ is:

$$\Delta\lambda = \lambda * \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta d}{d}\right)^2 + \left(\frac{\Delta L}{L}\right)^2}$$

$$\frac{\Delta x}{x} = \frac{0.0001}{0.0085} = 0.01176$$

$$\frac{\Delta d}{d} = \frac{0.00001}{0.000250} = 0.04000$$

$$\frac{\Delta L}{L} = \frac{0.001}{3} = 0.000333$$

$$\Delta\lambda = 7.0833 * 10^{-7} * \sqrt{(0.01176)^2 + (0.04000)^2 + (0.000333)^2}$$

$$\lambda = 2.95 * 10^{-8} m = 29.5nm$$

So for this trial $\lambda = (708.3 \pm 29.5) \, nm$

Calculation 7: Measurement Error Propagation

For the red laser,

The experimental value = 664.06 nm

Known value = 650 nm

Difference = 14.06 nm

Using a red laser trial at L = 3 m, d = 0.50mm, n = 2, x = 0.78 cm

$\lambda = \frac{xd}{(nL)}$, solving for Δx

$$\Delta x = \frac{(\Delta\lambda * nL)}{d}$$

$$\Delta x = \frac{(14.06*10^{-9} m * 2 * 3\, m)}{0.0005\, m}$$

$$\Delta x = 1.69 * 10^{-4} m = 0.169\, mm$$

This value shows an error of ~ 0.17 mm in the marking of the band positions that accounts for the 14.06 nm difference between the experimental average and the true value.

RESULTS AND DISCUSSION

The double-slit experiment successfully demonstrated both the wave and particle nature of light and provided a method to determine the wavelength of two different laser sources. The resulting experimental wavelength for the red laser was determined to be 664.06 nm with a standard deviation of 56.46 nm. When comparing this to the true value of the laser (650 nm), the percent difference was 2.16%. The experimental wavelength for the green laser was determined to be 531.25 nm and had a standard deviation of 32.10. When compared to the wavelength of the green laser (532 nm), the percent difference was only 0.14%.

The green laser's experimental average was significantly better when compared to the respective true value. Since the green laser has a shorter wavelength, the bands were closer together, which likely contributed to the identification and measurement of the bands.

The error propagation analysis for the red laser at 3 m revealed an uncertainty $\pm\ 29.5\, nm$ in the wavelength measurement. According to the error propagation, the dominant source of error was the uncertainty in the slit ($\frac{\Delta d}{d}$ = 0.04000), followed by the uncertainty when measuring the band positions ($\frac{\Delta x}{x}$ = 0.01176), and then likely negligibly the distance to the viewing screen ($\frac{\Delta L}{L}$ = 0.000333). To improve the accuracy of the experiment, decreasing the uncertainty of the slit would be most effective.

The 14.06 nm discrepancy between the experimental average and the true value for the red laser is a result of the measurement error of 0.17 mm in the marking of the fringe position.

Considering the measurements were taken with a ruler, this is reasonable precision given the difficulty of identifying the center of the bands.

The data at 2 m distance showed greater variability when compared to the 3 m trials. At shorter distances, the bands are closer together, indicating the importance of taking multiple measurements at different distances.

The calculation of the λ values, when compared to the true values, validates the double-slit experiment. Despite the small discrepancies, which can be explained due to experimental variables, the double-slit interference pattern can be used as a method for wavelength determination. Some potential improvements include using a more precise method of measuring the distances between the experimental variables, such as using computer software like Fiji, a program that allows for distances to be standardized with a picture much more precisely than a ruler.

REFERENCES

"27.3: Young's Double Slit Experiment." *Physics LibreTexts*, Libretexts, 1 Oct. 2024, phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_(OpenStax)/27%3A_Wave_Optics/27.03%3A_Youngs_Double_Slit_Experiment.

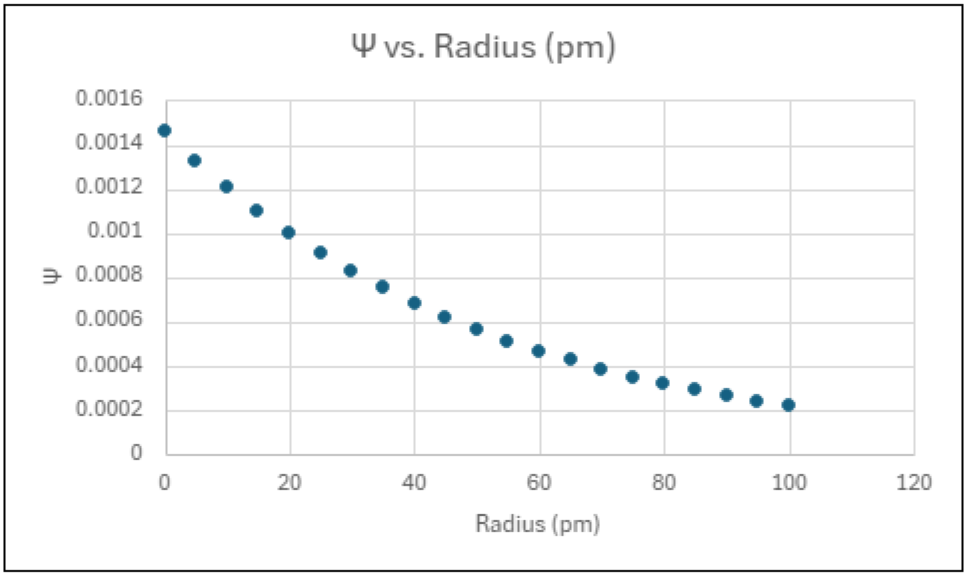
East Carolina University, Department of Chemistry. (2022). Physical Chemistry II Laboratory CHEM 3961 (Version 5.0) [Laboratory manual].

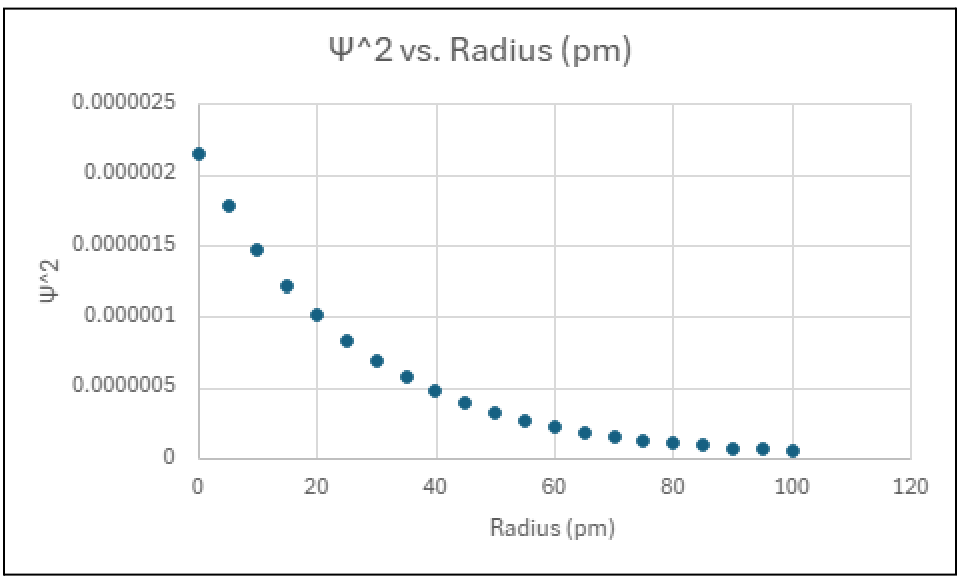
"Physics in a Minute: The Double Slit Experiment." *Plus Maths*, 19 Nov. 1970, plus.maths.org/content/physics-minute-double-slit-experiment.

APPENDICES

1. Atomic Orbitals Assignment:

Excel plots of the hydrogen atom wavefunction



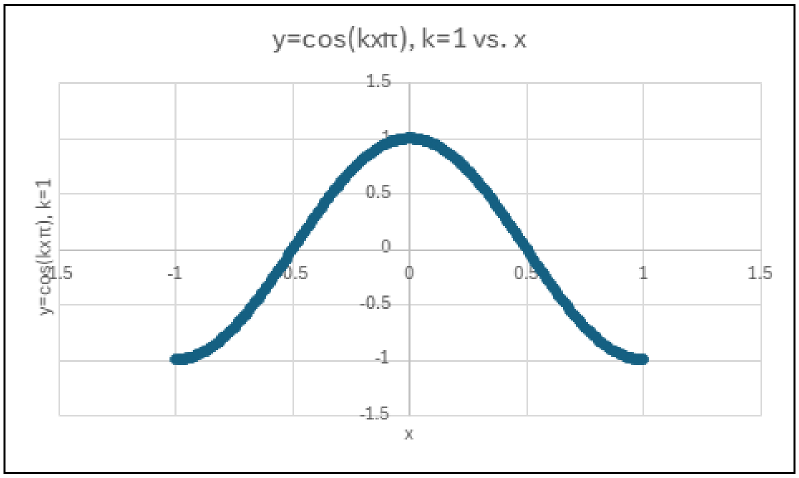


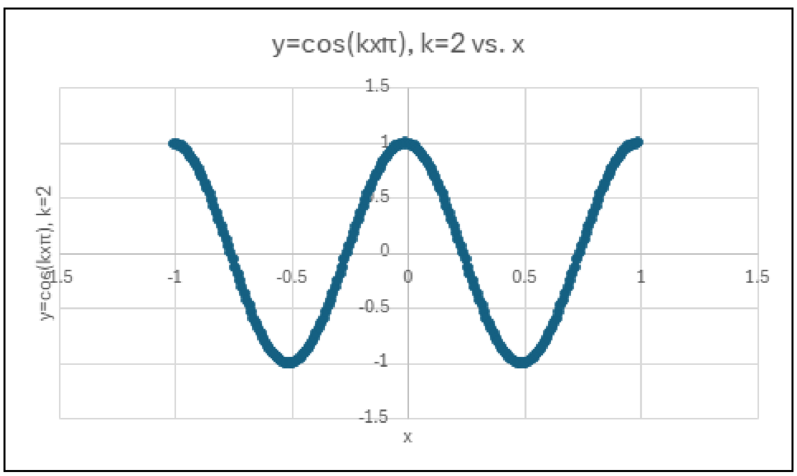
The wavefunction ψ is a mathematical representation of the quantum state of an electron in an atom. It can be positive, negative, or a complex value. It represents the amplitude of a probability wave.

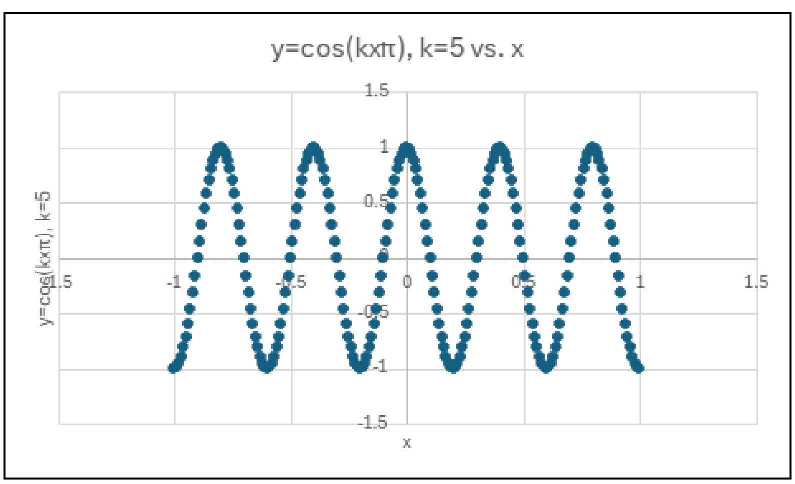
The normalized wavefunction, ψ^2, represents the probability density of finding an electron at a location in space. This value will always be positive and provides more meaningful information.

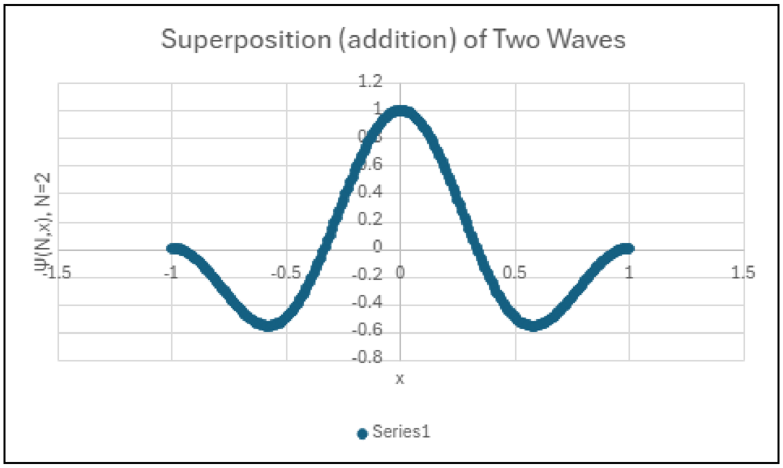
It tells you the probability of finding an electron at a particular location relative to the distance from the nucleus.

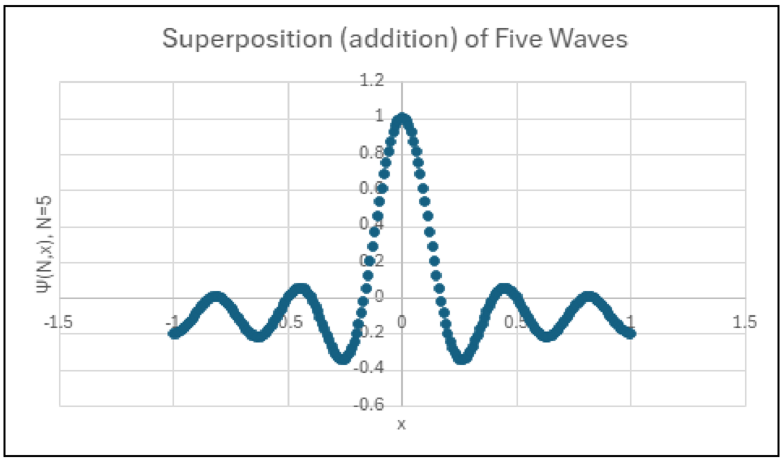
2. Superposition of Waves Assignment

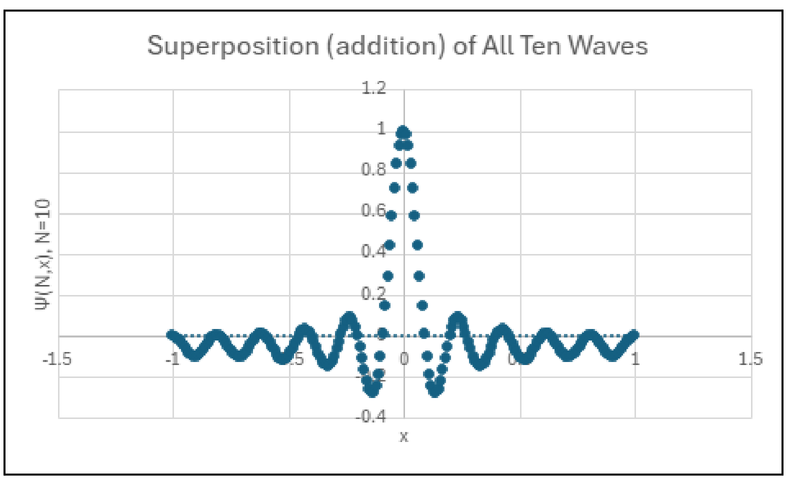







Superposition (addition) of Two Waves
Ψ(N,x), N=2
x
Series1


Superposition (addition) of Five Waves
Ψ(N,x), N=5
x


Superposition (addition) of All Ten Waves
Ψ(N,x), N=10
x